

Playwright Capstone Project

Project Title:

Enterprise E-commerce Automation Framework using Playwright with JavaScript (Node.js)

Website Name: nopCommerce Demo Store

Website URL: <https://demo.nopcommerce.com/>

Project Objective

This capstone project focuses on building a scalable and maintainable end-to-end automation framework using Playwright and JavaScript (Node.js). The project automates critical e-commerce workflows of the nopCommerce Demo Store platform while following enterprise automation best practices such as Page Object Model (POM), reusable utilities, data-driven testing, API validations, reporting, screenshots, and parallel execution.

Modules / Services Covered

1. **Authentication Management** – Login, logout, invalid login, forgot password, session validations.
2. **Product Management** – Product search, category filters, product sorting, product details validation.
3. **Cart Management** – Add/remove products, quantity update, cart persistence, cart validations.
4. **Checkout & Payment Flow** – Checkout process, address forms, payment validations, order confirmation.
5. **Compare & Order Management** – Product comparison, order placement, order history, invoice validation.
6. **User Profile Management** – Account details, password updates, address management, account deletion.
7. **Navigation & UI Validation** – Header/footer links, navigation menus, banners, responsive validations.
8. **API Testing & Mocking** – API request validation, response verification, route mocking, network interception.
9. **Search & Filter Validation** – Global product search, advanced filters, sorting options, result validations.

Technology Stack

- Playwright
- JavaScript (Node.js)
- VS Code
- HTML Reports
- GitHub Version Control